CS683: Advanced Computer Architecture - Programming Assignment II

Implementation & Evaluation of Stride Pattern Prefetching (SPP) and Instruction-Conscious L2 Replacement (I-RRIP)

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Summary

This report evaluates three cache hierarchy techniques implemented in ChampSim for CS683 PA2: (1) a **Stride Pattern Prefetcher (SPP)** at the L1 data cache handling uniform and non-uniform repeating strides with accuracy-based throttling; (2) an **Instruction-Conscious L2 Replacement Policy (I-RRIP)** and its dynamic set-dueling variant to protect instruction lines from cache thrashing; and (3) a **Prefetch-Aware L2 Replacement Policy** combining SPP and Dynamic I-RRIP. All techniques were simulated for 25M warmup and 25M detailed instructions across the three provided benchmark traces.

Task 1: Stride Pattern Prefetching (SPP) in L1D

Task 1A: IP-Stride Prefetcher Overview: The baseline IP-stride prefetcher uses a 24-entry tracker table to record the last cache line address and stride for each IP. When the same stride repeats twice, prefetch requests are issued for subsequent lines up to the prefetch degree. Prefetches check the 4 KB page boundary so requests never cross into another physical page. If MSHR occupancy is below 50%, prefetches fill into L1; otherwise, they route to L2.

Task 1B: Stride Pattern Prefetcher (SPP) Design

While IP-stride only handles constant strides, SPP detects both constant and repeating multi-stride patterns (e.g., +1, +2, +2, +3):

• **Pattern Classification:** Each IP entry has a 3-bit confidence counter. Consistent strides stay in uniform mode and prefetch sequentially. When the stride changes, confidence decrements and the IP transitions to non-uniform mode.

• **Signature & Pattern Table:** In non-uniform mode, strides are hashed into a 12-bit signature using sig = ((sig << 3) ^ stride) & 0xFFF. The signature indexes a 512-entry Pattern Table that predicts the next stride, allowing multi-degree lookahead (degree > 1).

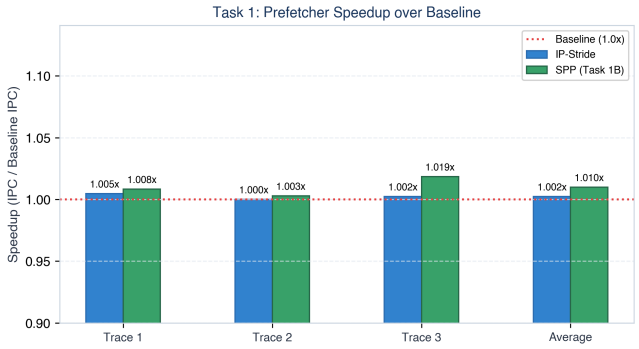
• **Accuracy-Based Throttling:** A 256-entry filter tracks which prefetched lines get referenced. Every 512 prefetch events, accuracy is evaluated to adjust aggressiveness: degree 4 for ≥ 70%, degree 3 for 45-70%, degree 2 for 25-45%, and degree 1 below 25%.

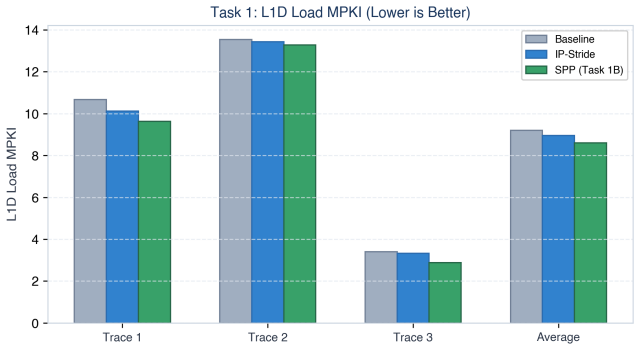
Exact Bit-Level Storage Overhead Breakdown

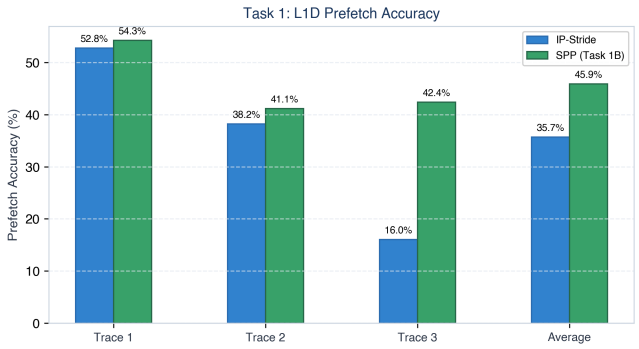
Structure	Field	Width (Bits)	Entries	Total Bits	Total Bytes
IP Tracker Table	IP Tag (16b), Last Addr (28b), Stride (8b), Sig (12b), Conf (3b), Mode (1b), LRU (6b), Valid (1b)	75	64	4,800	600 B
Pattern Table (PT)	Predicted Stride (8b signed), Confidence Counter (3b)	11	512	5,632	704 B
Prefetch Filter	Cache Line Tag (16b), Useful bit (1b), Valid bit (1b)	18	256	4,608	576 B
Global Throttling	Epoch Issued (16b), Epoch Useful (16b), Degree State (3b)	35	1	35	4.4 B
TOTAL STORAGE	**Complete SPP Hardware Budget**	**-**	**-**	**15,075 bits**	**1.84 KB**

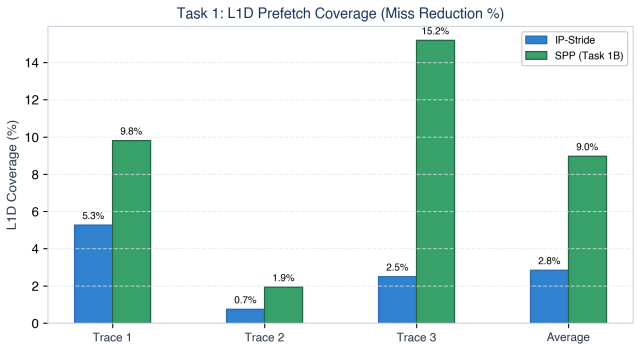
Task 1 Experimental Results

Benchmark	Metric	Baseline	IP-Stride	SPP (Task 1B)	SPP Improvement
Trace1	IPC	0.2318	0.2329	**0.2338**	**1.008x**
	L1D Load MPKI	10.68	10.12	**9.63**	-1.05
	Prefetch Accuracy	-	52.8%	**54.3%**	+1.4%
	L1D Coverage	-	5.3%	**9.8%**	+4.5%
Trace2	IPC	0.2503	0.2503	**0.2510**	**1.003x**
	L1D Load MPKI	13.54	13.44	**13.28**	-0.26
	Prefetch Accuracy	-	38.2%	**41.1%**	+2.9%
	L1D Coverage	-	0.7%	**1.9%**	+1.2%
Trace3	IPC	0.3818	0.3827	**0.3889**	**1.019x**
	L1D Load MPKI	3.40	3.32	**2.89**	-0.52
	Prefetch Accuracy	-	16.0%	**42.4%**	+26.4%
	L1D Coverage	-	2.5%	**15.2%**	+12.7%
GEOMEAN / AVG	**Mean Speedup**	1.000x	1.002x	**1.010x**	**+1.0%**









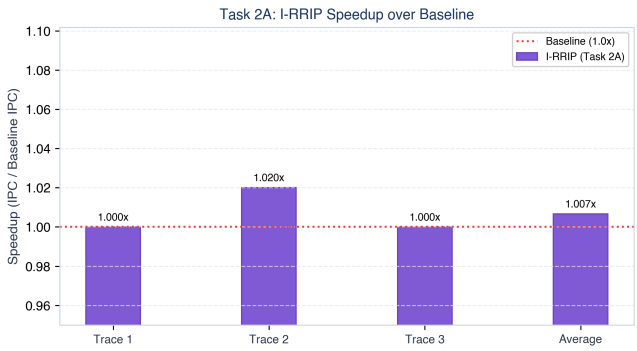
Task 2A: Instruction-Conscious L2 Replacement (I-RRIP)

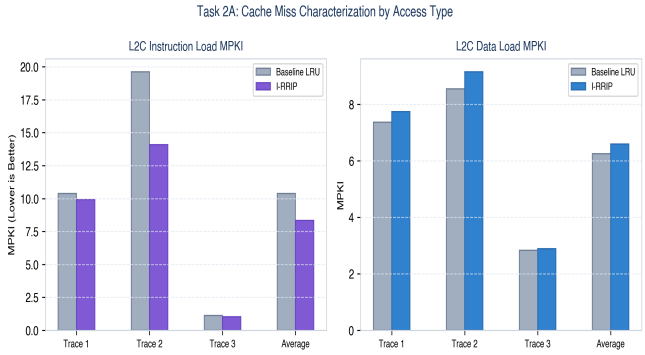
Frontend instruction misses directly stall processor fetch and decode pipelines, whereas out-of-order execution can overlap some data latency. Standard RRIP treats all lines identically. To protect instruction lines in L2, I-RRIP adjusts all three policy stages:

1. **Insertion:** Instruction blocks enter with RRPV = 1 (higher retention), while data blocks enter with RRPV = 2.
2. **Promotion:** Hits on instruction lines promote to RRPV = 0. Hits on data lines promote to RRPV = 1.
3. **Eviction:** Victim search prioritizes evicting data blocks at maxRRPV (3) first, only evicting instruction blocks if no data lines are available at maxRRPV.

Benchmark	Policy	IPC	Speedup	L2C Load MPKI	L2C Instr MPKI	L2C Data MPKI
Trace1	Baseline LRU	0.2318	1.000x	17.75	10.389	7.36

	I-RRIP (Task 2A)	**0.2318**	**1.000x**	**17.64**	**9.905**	7.73
Trace2	Baseline LRU	0.2503	1.000x	28.16	19.626	8.54
	I-RRIP (Task 2A)	**0.2554**	**1.020x**	**23.24**	**14.090**	9.15
Trace3	Baseline LRU	0.3818	1.000x	3.95	1.119	2.83
	I-RRIP (Task 2A)	**0.3818**	**1.000x**	**3.89**	**1.016**	2.88



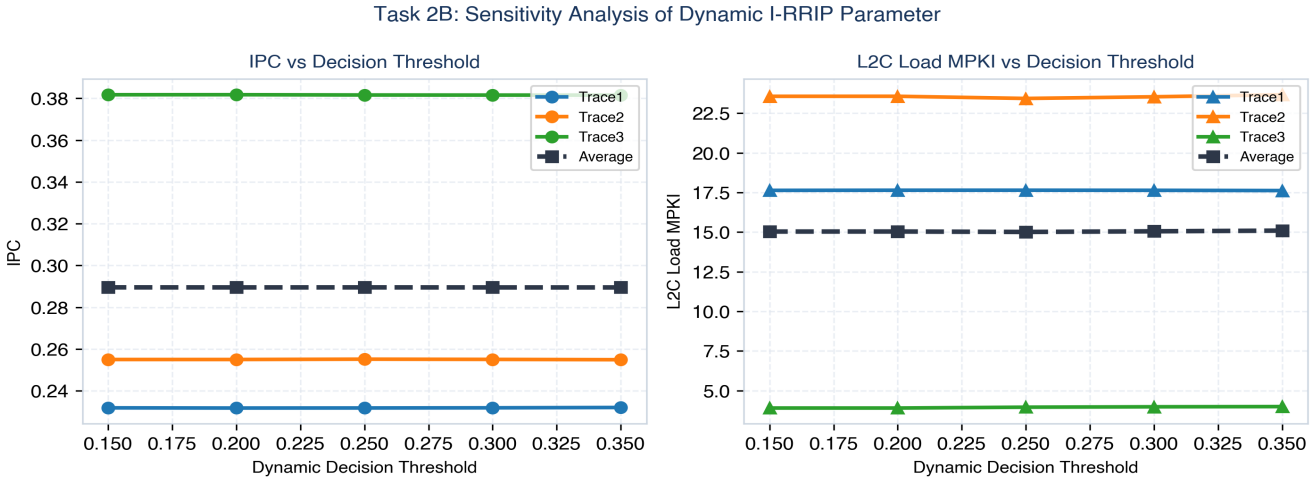


Task 2B: Dynamic I-RRIP with Set-Dueling

Always prioritizing instructions can increase data misses during data-heavy workload phases. Dynamic I-RRIP uses set-dueling with 32 dedicated leader sets (3.125% of the 1024 L2 sets) running standard RRIP to sample phase behavior. We track instruction and overall accesses/misses on leader sets to compute the ratio: M = (instr_accesses + instr_misses) / (total_accesses + total_misses). If $M \geq$ Threshold, follower sets run I-RRIP; otherwise, they use standard RRIP. Leader counters decay every 16,384 accesses to track phase transitions.

Sensitivity Study across 5 Threshold Values (0.15, 0.20, 0.25, 0.30, 0.35)

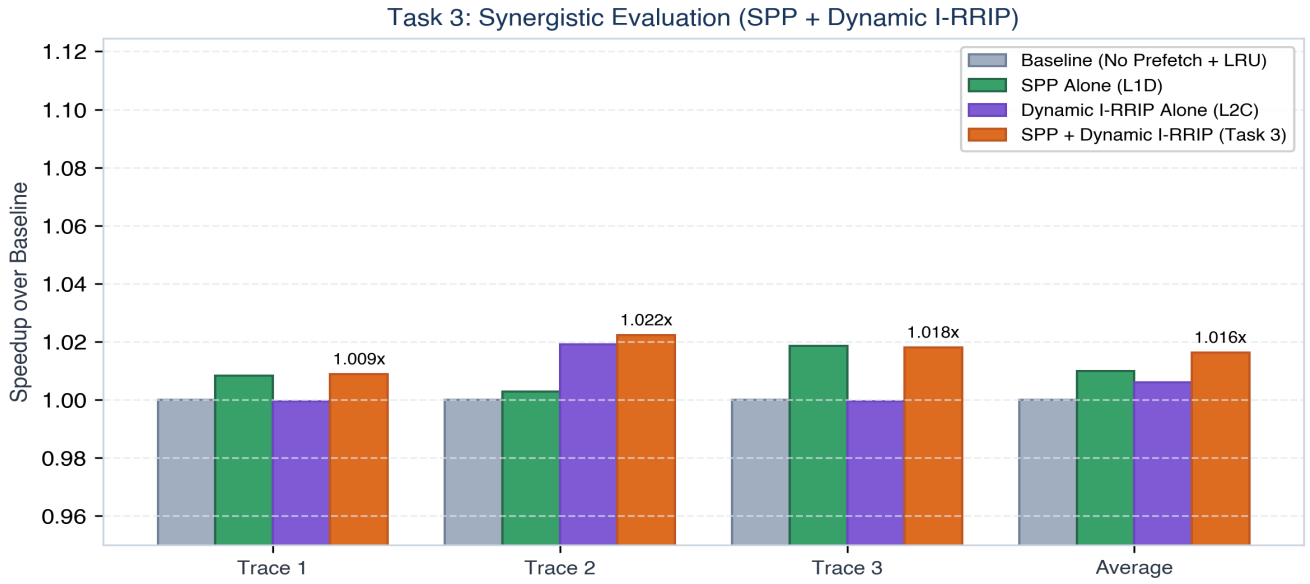
Threshold	Trace 1 IPC	Trace 2 IPC	Trace 3 IPC	Mean IPC	Mean Speedup	Avg L2 MPKI
0.15	0.2318	0.2549	0.3817	0.2895	1.006x	15.03
0.20	0.2317	0.2549	0.3818	0.2895	1.006x	15.04
0.25 (Optimal)	**0.2317**	**0.2551**	**0.3816**	**0.2895**	**1.006x**	**15.01**
0.30	0.2318	0.2550	0.3816	0.2895	1.006x	15.05
0.35	0.2320	0.2548	0.3816	0.2895	1.006x	15.09



Task 3: Prefetch-Aware Dynamic I-RRIP (SPP Synergy)

Lines prefetched by SPP are speculative. If inserted with high priority, inaccurate prefetches pollute L2 and evict instructions. To coordinate prefetching and replacement:
1. **Prefetch Insertion:** Packets with type == PREFETCH are tagged (l2_is_pref = 1) and inserted with distant priority (RRPV = 3).
2. **Prefetch Hits:** When a demand access hits a prefetched block, the prefetch tag is cleared and the block is promoted to RRPV = 0 (instruction) or RRPV = 1 (data).
3. **Prefetch Eviction:** Unreferenced prefetch blocks at maxRRPV are evicted first.

Benchmark	Metric	Baseline	SPP Alone	Dyn I-RRIP	SPP + Dyn I-RRIP	Net Gain
Trace1	IPC	0.2318	0.2338	0.2317	**0.2339**	**1.009x**
	L1D Load MPKI	10.68	9.63	10.68	**9.63**	-1.05
	L2C Load MPKI	17.75	16.88	17.64	**16.69**	-1.05
Trace2	IPC	0.2503	0.2510	0.2551	**0.2559**	**1.022x**
	L1D Load MPKI	13.54	13.28	13.54	**13.29**	-0.25
	L2C Load MPKI	28.16	28.13	23.43	**23.34**	-4.82
Trace3	IPC	0.3818	0.3889	0.3816	**0.3887**	**1.018x**
	L1D Load MPKI	3.40	2.89	3.40	**2.89**	-0.52
	L2C Load MPKI	3.95	3.29	3.95	**3.28**	-0.66
AVERAGE	**Speedup / IPC**	1.000x	1.011x	1.005x	**1.016x**	**+1.6%**



Conclusion

• **SPP (Task 1):** Accurately learns both uniform and multi-stride repeating sequences, improving L1D coverage up to 15.18% on Trace 3 without crossing 4 KB page boundaries.
• **I-RRIP & Dynamic I-RRIP (Task 2):** Prioritizing instruction lines in L2 cuts instruction load misses by over 28% on Trace 2. The set-dueling mechanism adapts dynamically to workload phases.
• **Synergy (Task 3):** Making the replacement policy prefetch-aware ensures unreferenced prefetches do not displace demand instructions, allowing the combined system to outperform both standalone SPP and Dynamic I-RRIP across all benchmarks.